

Perturbation-Conditioned Ecological Unbalanced Transport Formulation for Longitudinal Perturb-seq

Abstract

This note gives a self-contained formulation of a perturbation-conditioned ecological unbalanced transport model for longitudinal Perturb-seq with asymmetric observations. The experimental design motivates two coupled but non-identical components: an abundance model using matched pre-injection library measurements at T_0 , and a state-transport model using guide-labeled single-cell transcriptomic snapshots at P4 and P60. We introduce the full absolute population density, derive the induced equations for abundance and conditional state densities, show how the relative-abundance dynamics reduce to a replicator equation, and then incorporate ecological frequency dependence through a low-rank programme-level payoff acting primarily in the growth channel. The document is written pedagogically: every major quantity is defined explicitly, each equation is interpreted in words, and the modelling choices are justified from basic principles before optional extensions are introduced.

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1 Introduction and problem statement

1.1 What problem is being modelled?

We consider a longitudinal Perturb-seq setting in which the available measurements are not symmetric across time. At the pre-injection stage T_0 , we observe matched guide-library composition, typically as bulk guide counts or proportions. At later time points P4 and P60, we observe guide-labeled single-cell transcriptomic snapshots. Thus, the data contain

- an *exposure measurement* at T_0 that tells us how much of each perturbation entered the experiment, and

- *state-resolved single-cell measurements* at P4 and P60 that tell us which transcriptomic states are occupied by cells carrying each perturbation.

This distinction matters immediately. Because single-cell RNA sequencing is destructive, we never observe the same cell twice. Therefore, the temporal problem is fundamentally a *distributional* problem: one must explain how a population of cells observed at one time gives rise to a later population observed at another time. This is precisely the setting in which transport-based formulations are natural.

At the same time, simple mass-preserving transport is not sufficient. Different perturbations can expand, shrink, or disappear over time. In biological terms, cell populations proliferate, die, self-renew, differentiate, or lose competitive fitness. Hence, the relevant transport problem is *unbalanced*: mass is allowed to change.

Finally, the biological reading of the system suggests that perturbation fitness is not purely intrinsic. A perturbation can be favored or disfavored depending on the surrounding population composition and on mediator variables such as inflammatory programmes or microenvironmental context. For that reason, it is useful to augment transport with an *ecological* or *frequency-dependent* selection term.

1.2 Why the formulation must be asymmetric

A common first impulse is to write one state-space dynamical system from T_0 to P4 to P60. That would be appropriate only if transcriptomic cell states were observed at all three time points. Here they are not. The pre-injection time point T_0 provides only guide abundance information, not transcriptomic state measurements. Therefore, the clean formulation is:

1. abundance dynamics from $T_0 \rightarrow \text{P4} \rightarrow \text{P60}$;
2. state transport only from $\text{P4} \rightarrow \text{P60}$.

This does *not* make the model less joint. It remains joint because the same perturbation-conditioned quantities influence both abundance change and transcriptomic redistribution. What changes is that the jointness is *asymmetric*: abundance is modelled on the full interval, while state transport is only imposed where transcriptomic states are actually observed.

1.3 High-level modelling goal

The goal is to learn a perturbation-conditioned dynamical system that explains, at the same time,

- which perturbations become enriched or depleted relative to the matched input library,
- how cells move in latent transcriptomic space from P4 to P60 under each perturbation,
- how selection reshapes the internal distribution of states within each perturbation, and
- how the ecological composition of the full population modulates perturbation fitness.

The core idea is that abundance change and state redistribution should not be estimated as unrelated post hoc summaries. Instead, both should be consequences of the same perturbation-conditioned dynamical law.

2 Basic mathematical objects and notation

2.1 Indices, spaces, and times

We write

- $g \in \mathcal{G}$ for a perturbation index. In the main formulation, g is best interpreted at the *gene* level, with sgRNAs treated as technical or hierarchical replicates of the targeted gene.
- $x \in \mathbb{R}^d$ for a latent transcriptomic state. In practice, x can be obtained from PCA, a variational latent space, or another low-dimensional representation of the single-cell transcriptome.
- t for continuous time. We regard the state-resolved interval as $t \in [4, 60]$, corresponding to P4 and P60.

We will also use $u \in \{4, 60\}$ when writing discrete observation models at the measured time points.

2.2 Population-level decomposition

The cleanest way to formulate the model is to begin with the *absolute perturbation-specific state density*

$$\rho_g(x, t) \geq 0, \quad (1)$$

which represents the amount of population mass carried by perturbation g near latent state x at time t .

From ρ_g , define the *absolute abundance*

$$n_g(t) = \int_{\mathbb{R}^d} \rho_g(x, t) \, dx, \quad (2)$$

which is the total mass or effective population size of perturbation g .

Next define the *conditional state density within perturbation g*

$$p_g(x, t) = \frac{\rho_g(x, t)}{n_g(t)}, \quad \int_{\mathbb{R}^d} p_g(x, t) \, dx = 1. \quad (3)$$

This quantity describes the *shape* of the state distribution for perturbation g , independent of how many cells of that perturbation are present overall.

Finally define the *relative abundance*

$$f_g(t) = \frac{n_g(t)}{\sum_{h \in \mathcal{G}} n_h(t)}. \quad (4)$$

This is the perturbation frequency, namely the fraction of all perturbed cells that belong to perturbation g .

Combining these definitions gives

$$\rho_g(x, t) = n_g(t) p_g(x, t), \quad n_g(t) = N(t) f_g(t), \quad N(t) = \sum_{h \in \mathcal{G}} n_h(t). \quad (5)$$

This decomposition is conceptually central:

- $n_g(t)$ tells us how much of perturbation g is present in total,
- $p_g(x, t)$ tells us where that perturbation lives in state space,
- $f_g(t)$ tells us how common that perturbation is relative to all others,
- $\rho_g(x, t)$ combines abundance and state composition.

2.3 A notation table

Symbol	Meaning
$\rho_g(x, t)$	absolute perturbation-specific state density
$n_g(t)$	absolute abundance of perturbation g
$p_g(x, t)$	conditional state density within perturbation g
$f_g(t)$	relative abundance of perturbation g
$v_g(x, t)$	perturbation-conditioned drift field in latent state space
σ	diffusion scale (possibly replaced by state-dependent diffusion in extensions)
$r_g(x, t)$	local fitness or local growth/depletion field
$\bar{r}_g(t)$	perturbation-averaged fitness, obtained by integrating r_g against p_g
$\ell_{g,b}$	matched T_0 library frequency of perturbation g in batch b
$\zeta_g(t)$	log-enrichment relative to matched input library
$\eta(x)$	soft assignment of state x to ecological programmes
$q(t)$	population-level ecological composition
m_t	mediator vector summarizing contextual variables (for example programme scores or microenvironmental covariates)
a_g	low-dimensional embedding of perturbation g

3 State-resolved dynamics on the interval $[4, 60]$

3.1 Full absolute-density equation

On the state-resolved interval $t \in [4, 60]$, we model the absolute density ρ_g by the transport–diffusion–growth equation

$$\partial_t \rho_g(x, t) + \nabla_x \cdot (\rho_g(x, t) v_g(x, t)) = \frac{\sigma^2}{2} \Delta_x \rho_g(x, t) + r_g(x, t) \rho_g(x, t). \quad (6)$$

This single equation is the backbone of the formulation.

Interpretation of each term.

- $\partial_t \rho_g$ measures how the perturbation-specific population density changes over time.
- $\nabla_x \cdot (\rho_g v_g)$ is the transport term. The drift field $v_g(x, t)$ determines the directed motion of cells in latent state space.
- $\frac{\sigma^2}{2} \Delta_x \rho_g$ is the diffusion term. It captures stochastic spread or unresolved variability in cell-state dynamics.
- $r_g(x, t) \rho_g$ is the local growth or depletion term. Where r_g is positive, local mass expands; where r_g is negative, local mass shrinks.

Why this is an unbalanced transport model. Classical mass-preserving transport would omit the reaction term $r_g \rho_g$ and require total mass to be conserved. Here we intentionally allow mass to change, because perturbations can alter net population size through proliferation, survival, depletion, or competitive disadvantage. Thus, (6) is an *unbalanced* transport model.

3.2 Boundary information for the state equation

Because transcriptomic states are observed only at P4 and P60, the initial condition for state transport is taken from the empirical P4 guide-resolved single-cell distribution:

$$p_g(\cdot, 4) = \hat{p}_g^{(4)}(\cdot), \quad (7)$$

where $\hat{p}_g^{(4)}$ denotes the empirical or smoothed guide-conditioned density estimated from the P4 data.

No transcriptomic boundary condition is imposed at T_0 . This is not a modelling omission; it is a reflection of the experimental design.

4 Deriving abundance and within-perturbation state dynamics

The full PDE in (6) is useful, but it hides two qualitatively different phenomena:

1. the overall expansion or depletion of perturbation g ;
2. the internal reshaping of the state distribution within perturbation g .

The factorization $\rho_g = n_g p_g$ separates these phenomena cleanly.

4.1 Guide-averaged fitness

Define the perturbation-averaged fitness

$$\bar{r}_g(t) = \int_{\mathbb{R}^d} r_g(x, t) p_g(x, t) \, dx. \quad (8)$$

This is the average of the local fitness field $r_g(x, t)$ across the current state composition of perturbation g .

4.2 Absolute abundance dynamics

Integrating (6) over x and assuming standard no-flux boundary conditions or sufficiently fast decay at infinity yields the abundance equation.

Proposition 1 (Absolute abundance evolution). *Under the factorization $\rho_g = n_g p_g$ and suitable boundary conditions,*

$$\dot{n}_g(t) = n_g(t) \bar{r}_g(t). \quad (9)$$

Proof. Integrate (6) over $\mathbb{R}^d$:

$$\frac{d}{dt} \int \rho_g(x, t) \, dx + \int \nabla_x \cdot (\rho_g(x, t) v_g(x, t)) \, dx = \frac{\sigma^2}{2} \int \Delta_x \rho_g(x, t) \, dx + \int r_g(x, t) \rho_g(x, t) \, dx.$$

The divergence and Laplacian terms vanish under the boundary assumptions. Therefore,

$$\dot{n}_g(t) = \int r_g(x, t) \rho_g(x, t) \, dx.$$

Substitute $\rho_g(x, t) = n_g(t) p_g(x, t)$ to obtain

$$\dot{n}_g(t) = n_g(t) \int r_g(x, t) p_g(x, t) \, dx = n_g(t) \bar{r}_g(t).$$

□

Interpretation. Equation (9) says that the total abundance of perturbation g changes at its average fitness rate. If $\bar{r}_g(t) > 0$, perturbation g expands; if $\bar{r}_g(t) < 0$, it depletes.

4.3 Conditional state-density dynamics

Substituting $\rho_g = n_g p_g$ into (6) and using (9) yields the normalized state equation.

Proposition 2 (Conditional density evolution). *The within-perturbation state density obeys*

$$\partial_t p_g(x, t) + \nabla_x \cdot (p_g(x, t) v_g(x, t)) = \frac{\sigma^2}{2} \Delta_x p_g(x, t) + (r_g(x, t) - \bar{r}_g(t)) p_g(x, t). \quad (10)$$

Proof. Write (6) as

$$\partial_t (n_g p_g) + \nabla_x \cdot (n_g p_g v_g) = \frac{\sigma^2}{2} \Delta_x (n_g p_g) + r_g n_g p_g.$$

Because n_g depends only on t ,

$$\dot{n}_g p_g + n_g \partial_t p_g + n_g \nabla_x \cdot (p_g v_g) = \frac{\sigma^2}{2} n_g \Delta_x p_g + r_g n_g p_g.$$

Divide by n_g :

$$\frac{\dot{n}_g}{n_g} p_g + \partial_t p_g + \nabla_x \cdot (p_g v_g) = \frac{\sigma^2}{2} \Delta_x p_g + r_g p_g.$$

Using (9), we have $\dot{n}_g/n_g = \bar{r}_g$, which gives (10). □

Interpretation. Equation (10) is easy to misread, so its meaning should be stated carefully.

- The drift term still governs directed motion in state space.
- The diffusion term still governs stochastic spreading.
- The fitness contribution is now *centered*: $r_g(x, t) - \bar{r}_g(t)$.

The centering is essential because $p_g(\cdot, t)$ is a *conditional* density and must stay normalized. Therefore, only *relative fitness within perturbation g* can reshape p_g . The population-wide mean fitness $\bar{r}_g(t)$ is removed from the shape equation because it belongs to total abundance change, not internal redistribution.

Remark 1 (Separation of roles). *The pair of equations (9) and (10) produces a clean division of labour:*

$$\begin{aligned} \dot{n}_g(t) &= n_g(t) \bar{r}_g(t) && \text{describes total expansion or depletion,} \\ \partial_t p_g + \nabla \cdot (p_g v_g) &= \frac{\sigma^2}{2} \Delta p_g + (r_g - \bar{r}_g) p_g && \text{describes internal state reshaping.} \end{aligned}$$

This separation is one of the strongest advantages of the formulation.

5 Relative abundance and replicator dynamics

Absolute abundances $n_g(t)$ are useful, but in most perturbation screens the scientifically salient quantity is the relative abundance $f_g(t)$, because enrichment and depletion are inherently comparative.

5.1 Replicator equation

Define $N(t) = \sum_{h \in \mathcal{G}} n_h(t)$ and $f_g(t) = n_g(t)/N(t)$ as in (4). Then:

Proposition 3 (Replicator dynamics). *The relative abundance satisfies*

$$\dot{f}_g(t) = f_g(t) \left(\bar{r}_g(t) - \sum_{h \in \mathcal{G}} f_h(t) \bar{r}_h(t) \right). \quad (11)$$

Proof. Differentiate $f_g(t) = n_g(t)/N(t)$:

$$\dot{f}_g(t) = \frac{\dot{n}_g(t)}{N(t)} - \frac{n_g(t) \dot{N}(t)}{N(t)^2} = f_g(t) \left(\frac{\dot{n}_g(t)}{n_g(t)} - \frac{\dot{N}(t)}{N(t)} \right).$$

Now

$$\frac{\dot{n}_g(t)}{n_g(t)} = \bar{r}_g(t), \quad \frac{\dot{N}(t)}{N(t)} = \sum_{h \in \mathcal{G}} f_h(t) \bar{r}_h(t),$$

which gives (11). □

Interpretation. Equation (11) is the precise mathematical point where frequency dynamics become game-like. Perturbation g grows in *relative* abundance only if its average fitness exceeds the population-average fitness. If it does worse than average, its frequency decreases.

This is not a metaphor. It is exactly the classical replicator structure, now derived from the unbalanced transport formulation.

6 Observation model for matched-library-normalized guide abundance

6.1 Why T_0 enters as an offset

The matched library at T_0 tells us how much of each guide or gene entered the system before any in vivo selection occurred. Therefore, T_0 should appear as an *exposure offset*, not as a latent transcriptomic state.

Let $\ell_{g,b}$ denote the observed frequency of perturbation g in matched pre-injection library batch b . For sample s observed at time $u \in \{4, 60\}$ and matched to batch $b(s)$, define the expected perturbation fraction

$$\pi_{gs}(u) = \frac{\ell_{g,b(s)} \exp\{\zeta_g(u)\}}{\sum_{h \in \mathcal{G}} \ell_{h,b(s)} \exp\{\zeta_h(u)\}}. \quad (12)$$

Here $\zeta_g(u)$ is the log-enrichment or depletion of perturbation g relative to the matched input library.

Interpretation. Equation (12) separates two distinct sources of variation:

- $\ell_{g,b(s)}$: how much perturbation g was initially present in the experiment,
- $\zeta_g(u)$: how much perturbation g has expanded or depleted by time u relative to that starting amount.

This separation is important because a perturbation that is frequent at P60 might simply have been frequent already at T_0 , whereas a perturbation with modest final frequency may still be strongly enriched relative to a low initial exposure.

6.2 Count likelihood

Let $y_{\cdot s}(u)$ denote the observed vector of guide or perturbation counts in sample s at time u , and let N_s be the total number of guide-assigned cells. A convenient overdispersed likelihood is

$$y_{\cdot s}(u) \sim \text{DirMult}(N_s, \phi, \pi_{\cdot s}(u)), \quad (13)$$

where ϕ controls overdispersion.

Why use an overdispersed model? A multinomial likelihood assumes that once the expected proportions are fixed, all remaining variation is pure sampling noise. That is typically too restrictive for perturbation experiments, where additional variability arises from library preparation, capture efficiency, replicate-specific fluctuations, and compositional uncertainty. The Dirichlet–multinomial retains the multinomial mean structure while allowing extra dispersion.

6.3 Continuous-time interpretation of enrichment

Write the absolute abundance in the form

$$n_g(t) = \ell_g \exp\{\zeta_g(t)\}, \quad (14)$$

where ℓ_g denotes the matched input abundance reference for perturbation g .

Differentiating (14) yields

$$\dot{n}_g(t) = n_g(t) \dot{\zeta}_g(t).$$

Comparing with (9), we obtain

$$\dot{\zeta}_g(t) = \bar{r}_g(t). \quad (15)$$

Interpretation. The log-enrichment changes at the average fitness rate. Thus $\zeta_g(t)$ is not merely a statistical nuisance parameter; it is the time-integrated effect of the perturbation’s average fitness.

7 Ecological game: frequency-dependent selection through programme-level interactions

7.1 Why game theory enters here

The replicator equation (11) already shows that the model has the structure of frequency-dependent competition whenever the average fitnesses $\bar{r}_g(t)$ depend on the current population composition. In other words, game theory enters once fitness is made context dependent.

In this setting, context dependence should not be interpreted as direct pairwise combat between named perturbations. A more realistic biological interpretation is that perturbations alter how well cells in certain ecological programmes perform under the current population composition and mediator environment.

7.2 Programme-level strategies

Classical evolutionary games are formulated on a small set of strategies. Continuous transcriptomic state space $x \in \mathbb{R}^d$ is too high-dimensional to play that role directly. We therefore introduce a bridge from cell state to a small number of ecological programmes.

Let

$$\eta(x) \in \Delta^{K-1} \quad (16)$$

denote a soft assignment of state x to K ecological programmes, where Δ^{K-1} is the probability simplex. Thus, each coordinate $\eta_k(x)$ is nonnegative and $\sum_{k=1}^K \eta_k(x) = 1$.

Interpretation. A cell is not forced into a single hard class. Instead, $\eta(x)$ can encode partial membership in several programmes. This is especially useful when continuous developmental or inflammatory gradients blur sharp cell-state boundaries.

7.3 Population-level ecological composition

The global ecological composition is defined as

$$q(t) = \sum_{g \in \mathcal{G}} f_g(t) \int_{\mathbb{R}^d} \eta(x) p_g(x, t) \, dx. \quad (17)$$

The vector $q(t) \in \Delta^{K-1}$ describes which ecological programmes are currently abundant in the whole population.

Interpretation. Equation (17) first averages programme membership within each perturbation using p_g , then averages across perturbations using their relative abundances f_g . Thus $q(t)$ is a population-level ecological summary.

7.4 Mediator variables

Let

$$m_t \in \mathbb{R}^M \quad (18)$$

be a vector of mediator variables summarizing contextual quantities not fully captured by $q(t)$ alone. Depending on the application, m_t may represent programme scores, cell-type abundances, cytokine activity, or other microenvironmental features.

The point of m_t is not to create an arbitrary catch-all variable. Its role is to let the fitness landscape respond to relevant context that is measured or inferable from the data but is not fully specified by the programme mixture $q(t)$.

7.5 Drift parameterization

A conservative perturbation-conditioned drift parameterization is

$$v_g(x, t) = v_0(x, t) + u_\theta(x, a_g), \quad (19)$$

where

- $v_0(x, t)$ is a shared baseline drift field,
- $a_g \in \mathbb{R}^R$ is a low-dimensional embedding of perturbation g ,
- $u_\theta(x, a_g)$ is a perturbation-dependent deformation of the baseline drift.

This structure allows perturbations to alter the direction of movement in latent state space without introducing a separate unconstrained drift field for every perturbation.

7.6 Fitness parameterization

We write the local fitness as

$$r_g(x, t) = r_0(x, t) + \beta_g + \Phi(x, a_g, q(t), m_t), \quad (20)$$

where

- $r_0(x, t)$ is a shared state-dependent baseline fitness,
- β_g is a perturbation-specific state-independent fitness shift,
- Φ is an ecological interaction term.

Important consequence. Because β_g is independent of x , it contributes to $\bar{r}_g(t)$ and hence to abundance change, but it cancels inside $r_g(x, t) - \bar{r}_g(t)$ in the conditional state equation (10). Therefore, a constant perturbation-level advantage changes *how much* of perturbation g survives or expands, but does not by itself reshape the internal state composition.

7.7 Low-rank ecological payoff

The ecological payoff is defined by

$$\Phi(x, a_g, q, m) = \eta(x)^\top \left(P_0(m) + \sum_{r=1}^R a_{gr} P_r(m) \right) q, \quad (21)$$

where each $P_r(m) \in \mathbb{R}^{K \times K}$ is a programme-by-programme payoff matrix that may itself depend on the mediator state m .

Expanded entrywise,

$$\Phi(x, a_g, q, m) = \sum_{k=1}^K \sum_{\ell=1}^K \eta_k(x) \left[P_0(m) + \sum_{r=1}^R a_{gr} P_r(m) \right]_{k\ell} q_\ell. \quad (22)$$

Interpretation. Equation (22) has a direct ecological reading:

- $\eta_k(x)$ says how much the current cell behaves like programme k ,
- q_ℓ says how common programme ℓ is in the surrounding population,
- the matrix entry (k, ℓ) says how cells in programme k fare when programme ℓ is abundant,
- the perturbation embedding a_g modulates that interaction in a structured low-dimensional way.

7.8 Why the game is low-rank

A dense perturbation-by-perturbation interaction matrix is statistically inappropriate here. If there are $|\mathcal{G}|$ perturbations, then a fully dense pairwise interaction model already contains $|\mathcal{G}|^2$ coefficients before any state dependence is added. That is difficult to justify when only one state-resolved interval is observed.

The low-rank structure in (21) is not merely a convenience. It is a substantive assumption that many perturbations project onto a smaller number of ecological consequences. In applications motivated by convergent transcriptional programmes, that is usually the scientifically more plausible statement.

7.9 Why ecology should enter growth before drift

With only one transcriptomically resolved interval $P4 \rightarrow P60$, a fully flexible ecological effect in both drift and growth is weakly identifiable. A region of state space may become more populated because cells drifted there, because those cells preferentially grew there, or because both happened.

Abundance enrichment and depletion are observed more directly than full drift structure. For that reason, the most defensible core model puts ecology primarily into the fitness channel through Φ in (20), while keeping drift comparatively simpler through (19).

A more aggressive extension can be written as

$$v_g(x, t) = v_0(x, t) + u_\theta(x, a_g) + \lambda_v \nabla_x \Phi(x, a_g, q(t), m_t), \quad 0 < \lambda_v \ll 1, \quad (23)$$

but this should be treated as optional and strongly regularized.

8 Structural constraints

8.1 No guide switching

A cell does not change its perturbation label during the transport from P4 to P60. Therefore, the transport cost must be block-diagonal across perturbations. One convenient way to encode this is

$$c((x, g), (x', h)) = \|x - x'\|^2 + \infty \cdot \mathbf{1}\{g \neq h\}. \quad (24)$$

This means that state transport is allowed only within the same perturbation. Distinct perturbations interact only through the ecological composition $q(t)$ and the mediator variables m_t .

8.2 Absolute versus relative identifiability

Only *relative* abundance and *relative* fitness are identified.

First, adding the same constant to all perturbation log-enrichments does not change the normalized fractions in (12). Likewise, the replicator equation depends only on fitness differences relative to the population average.

Second, the low-rank factorization in (21) has an inherent scale ambiguity. Multiplying a_g by a constant and dividing the matrices $P_r(m)$ by the same constant leaves the product unchanged. Hence the model requires centering or scale constraints.

A convenient set of constraints is

$$\sum_{g \in \mathcal{C}} w_g \beta_g = 0, \quad (25)$$

for a designated control set $\mathcal{C}$, and

$$\frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} a_g = 0, \quad \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \|a_g\|^2 = 1. \quad (26)$$

8.3 Biological regime specificity

If one wishes to analyse a pre-malignant epithelial phase jointly with a later tumour phase, it is usually inappropriate to assume that a single stationary ecological payoff governs both. A safe extension is to let the baseline fields or payoff matrices be regime-specific:

$$P_r^{\text{pre}}(m), \quad P_r^{\text{tum}}(m), \quad (27)$$

or, more generally, to use a piecewise or regime-conditioned parameterization.

9 Training objective

9.1 Core loss

A convenient training objective is

$$\mathcal{L} = \mathcal{L}_{\text{count}} + \lambda_{\text{state}} \mathcal{L}_{\text{state}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{reg}} \mathcal{R}. \quad (28)$$

The individual terms are as follows.

Count loss. The abundance-fitting term matches observed guide or perturbation counts at P4 and P60 relative to the matched input library:

$$\mathcal{L}_{\text{count}} = \sum_{u \in \{4,60\}} \sum_s -\log p(y_{\cdot s}(u) \mid \pi_{\cdot s}(u)). \quad (29)$$

When using the Dirichlet–multinomial model, this is the negative log-likelihood induced by (13).

State loss. The state-fitting term compares the predicted P60 conditional distributions with the observed ones:

$$\mathcal{L}_{\text{state}} = \sum_{g \in \mathcal{G}} \text{Sink}_{\varepsilon}(\hat{p}_g^{(60)}, p_g(\cdot, 60)). \quad (30)$$

The entropic Sinkhorn divergence is a natural choice because it directly measures discrepancy between distributions in state space and is compatible with OT-style geometry.

Action regularization. To discourage unnecessarily complex dynamics, one can penalize the integrated magnitude of drift and growth:

$$\mathcal{L}_{\text{act}} = \sum_{g \in \mathcal{G}} \int_4^{60} \int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_g(x, t)\|^2 + \frac{\alpha}{2} |r_g(x, t)|^2 \right) \rho_g(x, t) \, dx \, dt. \quad (31)$$

This encourages the inferred dynamics to explain the data with minimal kinetic and growth complexity.

Auxiliary losses. The term $\mathcal{L}_{\text{aux}}$ may encode additional summary information such as programme scores, cell-type fractions, or other experimentally motivated statistics. These losses are optional; they should be used only when they correspond to measurements or summary quantities that the model is genuinely expected to reproduce.

Regularization. The term $\mathcal{R}$ collects generic penalties such as

- low-rank penalties on ecological parameters,
- smoothness penalties in x and t ,
- control-guide centering constraints,
- sgRNA-to-gene hierarchical shrinkage.

9.2 Interpretation of the training objective

The full loss in (28) has a simple high-level meaning.

- $\mathcal{L}_{\text{count}}$ fits abundance dynamics.
- $\mathcal{L}_{\text{state}}$ fits state redistribution.
- $\mathcal{L}_{\text{act}}$ prevents the model from using unnecessarily large drift and growth fields.
- $\mathcal{L}_{\text{aux}}$ injects additional experimentally observable structure, when available.
- $\mathcal{R}$ stabilizes parameters and improves identifiability.

Together, these terms ensure that abundance enrichment and transcriptomic state redistribution are explained by one perturbation-conditioned dynamical system rather than by disconnected downstream analyses.

10 Optional extensions

10.1 Abundance-only early phase from T_0 to P4

If one wishes to model the interval $[0, 4]$ more explicitly without inventing unobserved transcriptomic states, the defensible extension is an abundance-only early phase:

$$\dot{\zeta}_g(t) = \beta_g^E + a_g^\top B_E q^E(t), \quad t \in [0, 4], \quad (32)$$

where $q^E(t)$ is a low-dimensional ecological summary defined at the abundance level.

This extension preserves the asymmetry of the observed data: it enriches the abundance model before P4 without pretending that transcriptomic state trajectories were observed at T_0 .

10.2 Hierarchical sgRNA model

When several sgRNAs target the same gene, it is often preferable to share a gene-level core effect while allowing reagent-level deviations. Let j index sgRNAs nested within gene g . Then one may write

$$\beta_{gj} = \beta_g + b_{gj}, \quad b_{gj} \sim \mathcal{N}(0, \tau_\beta^2), \quad (33)$$

and

$$a_{gj} = a_g + \epsilon_{gj}, \quad \epsilon_{gj} \sim \mathcal{N}(0, \tau_a^2 I). \quad (34)$$

This treats sgRNAs as noisy replicates around a gene-level effect, which is usually more stable than fitting every guide fully independently.

10.3 State-dependent diffusion

The scalar diffusion coefficient σ in (6) is the simplest choice. A richer alternative is a state-dependent diffusion tensor $D_g(x, t)$:

$$\partial_t \rho_g + \nabla \cdot (\rho_g v_g) = \nabla \cdot (D_g(x, t) \nabla \rho_g) + r_g \rho_g. \quad (35)$$

This can capture heterogeneous stochasticity across regions of state space, but it also introduces additional identifiability burden. Therefore, the scalar model is a reasonable starting point unless the data strongly support more flexible diffusion.

11 Practical interpretation for implementation

To implement the model in practice, one may proceed in the following order.

1. Choose a latent transcriptomic space $x \in \mathbb{R}^d$ from the P4 and P60 single-cell data.
2. Aggregate sgRNAs to gene-level perturbations or define a hierarchical sgRNA model.
3. Estimate the empirical conditional distributions $\hat{p}_g^{(4)}$ and $\hat{p}_g^{(60)}$.
4. Construct matched-library offsets $\ell_{g,b}$ from the T_0 library data.
5. Define ecological programmes through an interpretable mapping $\eta(x)$.
6. Parameterize the baseline drift v_0 , perturbation drift shift u_θ , baseline fitness r_0 , and low-rank ecological payoff Φ .
7. Fit the model by minimizing (28) subject to the structural constraints and centering conditions.
8. Inspect both abundance-level outputs (ζ_g, f_g) and state-level outputs ($p_g(\cdot, t), q(t)$, programme contributions, and inferred fitness fields).

The conceptual sequence is equally simple:

T_0 gives exposure; P4 and P60 give state-resolved snapshots; transport explains where cells move; unbalanced growth explains how much each perturbation expands or depletes; and ecological payoffs explain why fitness depends on the current population composition and mediator context.

12 What the formulation does and does not claim

This note is deliberately about formulation rather than about a maximal novelty claim. The model should be interpreted as a *specific coupling* of several well-motivated ingredients:

- state transport for destructive snapshot data,
- unbalanced dynamics for changing population mass,
- matched-library normalization for perturbation abundance,
- frequency-dependent ecological fitness at the programme level.

What is important is not any one of these components in isolation, but the way they are tied together in an asymmetric Perturb-seq setting where the initial time point is abundance-resolved but not state-resolved.

13 Conclusion

The perturbation-conditioned ecological unbalanced transport model can be summarized in one sentence:

Use the matched T_0 guide library as an exposure offset for abundance dynamics, use P4 and P60 as state-resolved boundaries for transcriptomic transport, decompose the full perturbation-specific density into abundance and conditional state shape, and let a low-rank programme-level ecological payoff modulate perturbation fitness through the growth channel.

The reason this formulation is useful is that it keeps the modelling faithful to the data-generating process. It does not invent unmeasured transcriptomic states at T_0 ; it does not force mass conservation when clones obviously expand or contract; and it does not treat perturbation enrichment and state redistribution as unrelated summaries. Instead, one perturbation-conditioned dynamical system explains both abundance change and cell-state evolution, while still leaving room for ecological frequency dependence when the biology suggests it.

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